Program review

Example

1 issue group · 2 affected source lines

Repeated results are combined below. Every affected line is retained; different values, limits and calculation bases remain separate.

PROGRAM AT A GLANCE	
Saved software	5.24
Guidance	CB-Series / e-Series deployment guidance
Installation	Unavailable — installation settings omitted
Source coverage	4 script lines; 2 movel
Needs verification	15 grouped checks need runtime evidence or additional information
Generated	2026-11-02 18:17 UTC

Start here

Review the grouped issues on the next pages. Each entry gives the comparison, affected script lines and any source-estimate assumptions.

Source estimates substitute saved initializers; initialization and runtime values are not proven. UR acceleration recommendations are starting points, not safety limits. Unknown values are not treated as zero or a pass.

Issues to review

01 Acceleration recommendations

Configured a=4 m/s². Exceeds UR suggested starting value ≤2.5 m/s².

Lines 2–3
UR [1]
Example at line 2: movel(p[0,0,.2,0,0,0],a=4,v=.2,r=.01)

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Checks & source values

Issues already listed above are not repeated here. “Needs verification” means runtime evidence or a specification is missing. “Source value” records saved configuration, not physical correctness.

INITIAL PAYLOAD / COG / INERTIA	
Lines 1	Source value Mass: 2 kg; cog: [0, 0, 0.1] m from tool mount; inertia: [0, 0, 0, 0, 0, 0] kg·m²; transition_time: 0.3 s. UR [5]

DYNAMIC PAYLOAD CHANGES	
Program context	Needs verification 1 configuration call(s) in source, including definitions and branches. Call counts do not prove updates at every pickup, release or tool change. UR [1]

INITIAL TCP	
Program context	Needs verification No configuration call found in this source. Installation or external code may supply it. UR [1]

DYNAMIC TCP CHANGES	
Program context	Needs verification 0 configuration call(s) in source, including definitions and branches. Call counts do not prove updates at every pickup, release or tool change. UR [1]

Checks & source values (continued)

BLEND RADIUS OVERLAP	
Lines 2	Needs verification Configured radius r=0.01 m. Outgoing move distance=0.2 m. UR [4]
Lines 3	Needs verification Configured radius r=0.01 m. Incoming move distance=0.2 m. Adjacent radii total=0.02 m. UR [4]
Program context	Needs verification No explicit motion-version call found. Version 2 can reduce blend radii during planning. A static geometric overlap must not be presented as a proven skipped waypoint. UR [3]
Lines 2–3	Needs verification Configured blend radius=0.01 m (10 mm). The 2024 workshop suggests about 15 mm or more for general paths. Smaller radii can be intentional for clearance or process accuracy; review the actual path before increasing this value. This is not a safety limit or a proven motion fault. UR [12]

BLENDED MOTION CONTINUATION	
Program context	Needs verification 2 motion calls have nonzero or unresolved radii. Verify continuation through branches, waits, stops and helper functions on the applicable controller. Static source inspection does not establish the executed path. UR [4]

VELOCITY SPECIFICATIONS	
Lines 2–3	Needs verification Configured v=0.2 m/s. Enter the applicable TCP velocity specification to check this value. UR [1]

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Checks & source values (continued)

MOVE DISTANCES	
Lines 2	Needs verification Move distance unresolved: initial pose, joint-space trajectory, target/frame expression or intervening control flow is unknown. UR [4]
Lines 3	Source value Unblended move distance=0.2 m from preceding source waypoint (translation only). UR [4]
Program context	Source value Known source path distance=0.2 m across 1 of 2 move(s). This sums resolvable unblended translational segments only; unknown segments, first approach, rotations, branches and repetitions are excluded. It is not total executed travel. UR [4]

PAYLOAD TRANSITION AND INERTIA	
Lines 1	Needs verification Review the payload transition against the actual pickup/release. A nonzero transition can help with heavy parts or vacuum gripping; the program continues during the transition, so motion timing still needs validation. This call specifies 0.3 s. Zero/default inertia is present. Check whether a measured or CAD-derived inertia is needed for a long or widely distributed load; zero inertia is not automatically a fault. UR [11]

PROCESS MOTION AND WAITS	
Program context	Needs verification No MoveP call was found. Verify that waits, branches and continuously checked conditions do not interrupt blended motion. Use appropriate controlled stops where interruption is required; external helpers and executed paths are not resolved by this review. UR [13]

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Checks & source values (continued)

DIAGNOSTICS AND PROGRAM EFFICIENCY	
Lines 4	Needs verification 1 socket open/close source call(s). Review connection lifetime and reuse where practical. Repeated connection setup can burden the controller; source counts do not establish how often the calls execute. UR [13]
Program context	Needs verification Review thread/event load and busy loops for appropriate yielding (such as sync or a wait). A sync or sleep elsewhere in the file does not prove that every executed loop yields. Review unnecessary URCaps and interface activity if PolyScope is slow. UR [13]
Program context	Needs verification For supervised troubleshooting, consider position_deviation_warning(True, 0.4), the workshop starting point, or pause_on_error_code for the specific warning. Confirm support in the applicable controller manual. These optional diagnostics are not installed or executed by this report. UR [14]

Physical installation follow-up

Not assessed by this report. Source files and telemetry cannot verify these items. Record the inspection evidence and outstanding actions in the deployment checklist.

Manual checks · all unassessed	
Mounting and orientation	Record the base/tool mounting assessment, fastening inspection and configured orientation. Include moving-base conditions where applicable. Source: Guide p. 5; Workshop pp. 5–10
Environment and protection	Record suitability for the arm, controller and pendant: contamination, moisture, temperature, ventilation and cable routing. Source: Guide p. 5; Workshop pp. 12–18
Power and I/O	Record the electrical inspection reference, power quality and I/O load review. Use model-specific electrical limits. Source: Guide p. 9; Workshop p. 17
Risk assessment	Reference the application risk assessment and approved safety configuration, including tooling, workpieces and connected machinery. Do not store passwords here. Source: Guide p. 12
Safety validation	Reference the qualified validation record for emergency stops, safeguards, modes and connected safety I/O after installation or modification. Source: UR e-Series Service Manual, Pre-Use Assessment (SW5.24)
Inspection plan	Record the model-specific inspection plan, responsible person and next review date. Use the manual and operating conditions to set intervals. Source: Workshop pp. 54–56; UR e-Series Service Manual, Inspection Plan (SW5.24)

Guide: UR Installation Guide v1.3.3. Workshop: UR Application Deployment Workshop, May 2024. Page numbers refer to the supplied PDFs. Apply the current model-specific manual; these prompts are not a complete safety assessment.

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Source notes

How to interpret the review

This is a static source review; no URScript is executed. Counts include definitions and branches, not execution frequency. Source values do not establish measured speeds or accelerations. Blends depend on controller planning; a geometric warning does not prove a skipped waypoint. Validate physical payload, centre of gravity, inertia and TCP, including each workpiece or tool change.

Reviewed source files

/tmp/tmp9ex8wm5d/example.urp
SHA-256: 996a20b6db1c7b658475fa9eef001be7c03034e0d3dd92b7e3642bcbd15cd13a

/tmp/tmp9ex8wm5d/example.script
SHA-256: f9447487fc65928477d91a246a881c1cb4fa85965d14b59562358d5bcd351bbc

UR references

[1] UR Important Deployment Points (CB-Series/e-Series)
https://www.universal-robots.com/articles/ur/robot-care-maintenance/important-deployment-points/

[3] UR Motion Version (PolyScope 5.22+)
https://www.universal-robots.com/manuals/EN/HTML/SW5_22/Content/prod-scriptmanual/all_scripts/motion_version_desc.htm

[4] UR Blend Parameters
https://www.universal-robots.com/manuals/EN/HTML/SW5_19/Content/prod-usr-man/software/PolyScope/content/BasicProgNodes/commandtab_way_blend_para.htm

[5] UR Payload configuration (PolyScope 5.24)
https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-scriptmanual/all_scripts/set_target_payload.htm

[11] UR Application Deployment Workshop (May 2024), PDF pp. 25, 29–30: payload transitions and inertia
https://s3-eu-west-1.amazonaws.com/ur-support-site/100103/UR%20Application%20Deployment_Workshop_2024.pdf

[12] UR Application Deployment Workshop (May 2024), PDF pp. 33–36: blends and process motion
https://s3-eu-west-1.amazonaws.com/ur-support-site/100103/UR%20Application%20Deployment_Workshop_2024.pdf

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UR references (continued)

[13] UR Installation Guide v1.3.3, PDF pp. 7, 10–11 (printed pp. 6, 9–10): continuity, processing and interfaces
https://s3-eu-west-1.amazonaws.com/ur-support-site/100103/Installation%20Guide%20V1.3.3_ENGLISH.pdf

[14] UR Application Deployment Workshop (May 2024), PDF pp. 58–61: diagnostic instrumentation
https://s3-eu-west-1.amazonaws.com/ur-support-site/100103/UR%20Application%20Deployment_Workshop_2024.pdf

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